MD-like Actor-Critic Algorithms

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Consider the case of unparametrized policies:

$$\pi_\theta(a|s) = \pi(a|s)$$

Recall:

$$\nabla_\pi J_\rho = \frac{1}{1-\alpha} E_{s\sim \nu_\pi(\cdot)} \left(\pi(a|s) \nabla_\pi \log \pi(a|s) A_\pi(s,a) \right)$$

since $\frac{\partial}{\partial \pi(a'|s')} \pi(a|s) = \begin{cases} 1 & \text{if } (a',s') = (a,s) \\ 0 & \text{else} \end{cases}$

$$\Rightarrow \left(\nabla_\pi J_\rho\right)[s,a] = \frac{1}{(1-\alpha)} \nu_\pi(s) A_\pi(s,a)$$

Consider a mirror descent type algorithm: given π_k, π_{k+1} is obtained by solving

$$\max_{\pi:} \frac{1}{1-\alpha} \sum_{s,a} \nu_{\pi_k}(s) A_{\pi_k}(s,a) \left(\pi(a|s) - \pi_k(a|s)\right) - \frac{1}{\epsilon} \sum_s \nu_\pi(s) D_{KL}\left(\pi(\cdot|s) \| \pi_k(\cdot|s)\right)$$

$\sum_a \pi(a|s) = 1 \ \forall s$

$\pi(a|s) \geq 0 \ \forall s,a)$

The objective decomposes per state as

$$\max_{\sum_a \pi(a|s) = 1} \frac{1}{1-\alpha} \sum_a A_{\pi_k}(s,a) \left(\pi(s|a) - \pi_k(a|s)\right)$$

$$\max_{\pi} \quad \frac{1}{1-\alpha} \sum_a A_{\pi_k}(s,a)\left(\pi(s|a) - \pi_k(a|s)\right)$$

$$\sum_a \pi(a|s) = 1$$

$$\pi(a|s) \geq 0 \quad \forall a$$

$$- \frac{1}{\varepsilon} \sum_a \pi(a|s) \log \frac{\pi(a|s)}{\pi_k(a|s)}$$

Differentiating w.r.t. π

$$\frac{1}{1-\alpha} A_{\pi_k}(s,a) - \frac{1}{\varepsilon}\left(\log \frac{\pi(a|s)}{\pi_k(a|s)} - 1\right) = 0$$

$$\log \frac{\pi(a|s)}{\pi_k(a|s)} = \frac{\varepsilon}{1-\alpha} A_{\pi_k}(s,a) + 1$$

$$\pi_{k+1}(a|s) = \frac{\pi_k(a|s) \exp\left(\frac{\varepsilon}{1-\alpha} A_{\pi_k}(s,a)\right)}{Z_{k+1}(s)} \quad \forall s$$

$$Z_{k+1}(s) := \sum_{a'} \pi_k(a'|s) \exp\left(\frac{\varepsilon}{1-\alpha} A_{\pi_k}(s,a)\right)$$

Note: A_π can be estimated using TD-learning.
$\Rightarrow$ We have an actor-critic algorithm.

Next, we will prove the convergence of the algorithm assuming A_{π_k} is estimated exactly.

We first recall the performance difference lemma:

$$J_\rho(\pi) - J_\rho(\pi') = \frac{1}{1-\gamma} E_{(s,a) \sim \eta_\pi}\left(A_{\pi'}(s,a)\right)$$

$$J_\rho(\pi) - J_\rho(\pi') = \frac{1}{(1-\alpha)} E_{(s,a)\sim\eta_\pi}\left(A_{\pi'}(s,a)\right)$$

Next, consider the Lyapunov function

$$W(\pi_n) = \sum_s \eta_{\pi^*}(s)\, D_{KL}\left(\pi^*(\cdot|s) \,\|\, \pi_n(a|s)\right)$$

$$W_{\pi_{n+1}} - W_{\pi_n} = \sum_s \eta_{\pi^*}(s) \sum_a \pi^*(a|s) \log \frac{\pi_n(a|s)}{\pi_{n+1}(a|s)}$$

$$= \sum_s \eta_{\pi^*}(s) \sum_a \pi^*(a|s)\left(\log Z_{n+1}(s) - \frac{\varepsilon}{1-\alpha} A_{\pi_n}(s,a)\right)$$

$$= \sum_s \eta_{\pi^*}(s) \sum_a \pi^*(a|s) \log Z_{n+1}(s) - \varepsilon\left(J_\rho(\pi^*) - J_\rho(\pi_n)\right)$$

↑ Performance difference lemma

$$= \sum_s \eta_{\pi^*}(s) \sum_a \pi_{n+1}(a|s) \log Z_{n+1}(s) - \varepsilon\left(J_\rho(\pi^*) - J_\rho(\pi_n)\right)$$

↑ replace $\sum_a \pi^*(a|s)$ with $\sum_a \pi_{n+1}$ since both are $=1$

$$= \sum_s \eta_{\pi^*}(s) \sum_a \pi_{n+1}(a|s) \log \frac{\pi_n(a|s)}{\pi_{n+1}(a|s)}$$

$$= -D_{KL}\left(\pi_{n+1}(\cdot|s) \,\|\, \pi_n(\cdot|s)\right) \le 0$$

$$+ \sum_s \eta_{\pi^*}(s) \frac{\varepsilon}{1-\alpha} \sum_a \pi_{n+1}(a|s) A_{\pi_n}(s,a) - \varepsilon\left(J_\rho(\pi^*) - J_\rho(\pi_n)\right)$$

$$+ \sum_s \eta_{\pi^+}(s) \frac{\varepsilon}{1-\alpha} \sum_s \pi_{n+1}(a|s) A_{\pi_n}(s,a) - \varepsilon \left(J_\rho(\pi^+) - J_\rho(\pi_n) \right)$$

$$\leq \frac{\varepsilon}{1-\alpha} \sum_s \rho'(s) \pi_{n+1}(a|s) A_{\pi_n}(s,a) - \varepsilon \left(J_\rho(\pi^+) - J_\rho(\pi_n) \right)$$

$\uparrow$

$\rho'(s) \triangleq \eta_{\pi^+}(s)$

$$\leq \frac{\varepsilon}{(1-\alpha)^2} \sum_s \eta^{\rho'}_{\pi_{n+1}}(s) \pi_{n+1}(a|s) A_{\pi_n}(s,a) - \varepsilon \left(J_\rho(\pi^+) - J_\rho(\pi_n) \right)$$

$\uparrow$

Recall $\eta^{\rho'}_{\pi_{n+1}}$ is the state-visitation distribution under π_{n+1} starting from $\rho'(\cdot)$

$$\frac{1}{1-\alpha} \eta^{\rho'}_{\pi_{n+1}}(s) = \rho'(s) \left[1 + \alpha P(s_1 = s \mid s_0 = s) + \alpha^2 P(s_2 = s \mid s_0 = s) + \cdots \right]$$

$$\Rightarrow \eta^{\rho'}_{\pi_{n+1}}(s) \geq (1-\alpha) \rho'(s)$$

$$= \frac{\varepsilon}{1-\alpha} \left(J_{\rho'}(\pi_{n+1}) - J_{\rho'}(\pi_n) \right) - \varepsilon \left(J_\rho(\pi^*) - J_\rho(\pi_n) \right)$$

$\uparrow$

Perf. diff. lemma again

$$\Rightarrow J_\rho(\pi^+) - J_\rho(\pi_n) \leq \left(J_{\rho'}(\pi_{n+1}) - J_\rho(\pi_n) \right) \frac{1}{1-\alpha} + \frac{W(\pi_n) - W(\pi_{n+1})}{\varepsilon}$$

$\Rightarrow$

$$T(\pi^+) - \frac{1}{T} \sum_{t=1}^{T} J_\rho(\pi_n) \leq J_{\rho'}(\pi_T) - J_\rho(\pi_0) + W(\pi_0)$$

$$J_\rho(\pi^*) - \frac{1}{T}\sum_{k=0}^{T-1} J_\rho(\pi_k) \le \frac{J_{\rho'}(\pi_T) - J_\rho(\pi_0)}{(1-\alpha)T} + \frac{W(\pi_0)}{\varepsilon T}$$

$$\le \frac{2 r_{max}}{(1-\alpha)^2 T} + \frac{W(\pi_0)}{\varepsilon T}$$

Choose $\pi_0(a|s) = \frac{1}{|A|} \quad \forall s, a$, where A is the set of actions

$$\Rightarrow W(\pi_0) = \sum_s \eta_{\pi^*}(s) \sum_a \pi^*(a|s) \left(\log \pi^\alpha(a|s) + \log |A| \right)$$

$$\le \log |A|$$

$$\Rightarrow \boxed{J_\rho(\pi^*) - \frac{1}{T}\sum_{k=0}^{T-1} J_\rho(\pi_k) \le \frac{2 r_{max}}{(1-\alpha)^2 T} + \frac{\log |A|}{\varepsilon T}}$$

If we can now show $J_\rho(\pi_k) \ge J_\rho(\pi_{k-1}) \quad \forall k$

$$\Rightarrow \boxed{J_\rho(\pi^*) - J_\rho(\pi_{T-1}) \le \frac{2 r_{max}}{(1-\alpha)^2 T} + \frac{\log |A|}{\varepsilon T},}$$

thus completing the proof.

To show that $J_\rho(\pi_{n+1}) - J_\rho(\pi_n)$, we note that

$$J_\rho(\pi_{n+1}) - J_\rho(\pi_n) = \frac{1}{1-\alpha} \sum_{s,a} \eta^{\rho}_{\pi_{n+1}}(s,a)\, A_{\pi_n}(s,a)$$

$$\geq \frac{1}{(1-\alpha)^2} \sum_{s,a} \rho(s)\, \pi_{n+1}(a|s)\, A_{\pi_n}(s,a)$$

$$= \frac{1}{\varepsilon(1-\alpha)} \sum_{s,a} \rho(s)\, \pi_{n+1}(a|s) \left[\log \frac{\pi_{n+1}(a|s)\, Z_{n+1}(s)}{\pi_n(a|s)} \right]$$

$$\geq \frac{1}{\varepsilon(1-\alpha)} \sum_{s,a} \rho(s)\, \pi_{n+1}(a|s) \log Z_{n+1}(s)$$

$$= \frac{1}{\varepsilon(1-\alpha)} \sum_{s} \rho(s) \log Z_{n+1}(s)$$

$$\log Z_{n+1}(s) = \log \sum_a \pi_n(a|s) \exp\left(\frac{\varepsilon}{1-\alpha} A_n(s,a) \right)$$

$$\geq \sum_a \pi_n(a|s)\, \frac{\varepsilon}{1-\alpha} A_n(s,a)$$

Jensen's

$$= \sum_a \pi_n(a|s)\, \frac{\varepsilon}{1-\alpha} \left(Q_n(s,a) - V_n(s) \right)$$

$$= 0$$